

Rahul Madhav Deshpande

Baltimore, MD | (510) 953-2367 | rahulmadhavd04@gmail.com | linkedin.com/in/rahulmadhavdeshpande

EDUCATION

Johns Hopkins University - Whiting School of Engineering
M.S.E., Mechanical Engineering - Robotics and Intelligent Systems Track
Pennsylvania State University, University Park
B.S., Mechanical Engineering; Minor, Mechatronics

Baltimore, MD
Aug 2026 - Present
State College, PA
May 2026

WORK EXPERIENCE

Biophotonics and Ultrasound Imaging Lab, Penn State
Undergraduate Research Assistant

State College, PA
Aug 2025 - Jun 2026

- Spearheaded the transition of focused transparent ultrasound transducer (f-TUT) array fabrication from planar to focused geometries, and designed the flex PCB, interconnect, and matched-impedance board for the resulting 70-element array, improving photoacoustic imaging performance and device yield without degrading signal integrity.
- Developed and trained neural networks for photoacoustic image enhancement using paired photoacoustic/ultrasound data, validating generalization with NIRfast- and K-Wave-simulated tissue-mimicking phantoms.
- Conducted piezoelectric material analysis across multiple cuts of Lithium Niobate and PMN-PT samples, using KiCad simulations to derive electromechanical parameters informing next-generation transducer design decisions.

Siemens Centre of Excellence (GTTC)
Mechanical Engineering Intern

Mysore, India
Jul 2025 - Aug 2025

- Developed and validated robotic kinematics models in ROS 2, applying forward and inverse kinematics to optimize joint trajectory planning and improve industrial-arm path repeatability 22% through precise encoder calibration.
- Programmed CNC lathe and milling operations using Siemens Sinumerik to machine high-tolerance prototype components to ± 0.02 mm dimensional accuracy, cutting cycle time 15%.

NASA L'SPACE Mission Concept Academy
Mission Power Subsystem Lead

Remote, USA
Jan 2025 - Jun 2025

- Led design and trade studies for a Venus Aerobot power subsystem under a \$200M budget and 50 kg mass limit, delivering ≥ 200 W continuous power under 150°C, 92-bar conditions.
- Applied NASA Systems Engineering principles to author Verification & Validation and Science Traceability matrices, demonstrating systems-engineering rigor directly transferable to safety-critical medical device design.

Department of Mechanical Engineering, Penn State
Undergraduate Research Assistant

State College, PA
Feb 2024 - Dec 2024

- Developed LiDAR-based object-avoidance algorithms in Python and ROS 2, improving real-time obstacle-detection accuracy 20% over baseline Euclidean clustering methods.
- Integrated Crazyflie drones with a Vicon motion-capture system, using its ground-truth data to calibrate onboard perception sensors and achieve ± 2 cm localization accuracy.

CLUBS AND INVOLVEMENT

Penn State Hyperloop Development Team
President & Co-Founder

State College, PA
Aug 2024 - May 2026

- Co-founded and led a 30-member, 8-discipline team designing and prototyping a next-generation hyperloop pod, drafting the team constitution and building its sponsorship and partnerships strategy from the ground up.
- Directed development of a 12-slot dual-sided linear induction motor (DSLIM) and in-house battery management system, theoretically achieving over 68% thrust efficiency.

American Society of Mechanical Engineers (ASME) - PSU Student Chapter
Steering & Suspension Team Lead

State College, PA
Jan 2024 - May 2025

- Co-led an 8-member team in the design and development of TARMO5, an advanced RC vehicle, increasing cornering stability and speed 20% over previous iterations through linkage-geometry optimization validated in ANSYS and SolidWorks Simulation.

RELEVANT MEDICAL DEVICE PROJECTS

- **Biomechanical Tremor & Handwriting Analysis Pen** (*Medical Mechatronics & Human Systems Design*): Built a MEMS-based smart pen (accelerometer + force sensing) quantifying 4-12 Hz tremor and writing kinematics; developed a 30-feature ML pipeline that reached over 92% stress-classification accuracy.
- **Endotracheal Cuff Pressure Manometer** (*Medical Device Design Project*): Engineered a digital manometer (ESP32, piezoresistive sensing) monitoring 0-40 kPa cuff pressures to $< \pm 1\%$ full-scale accuracy, with OLED readout and wireless data logging enabling real-time surgical feedback without obstructing the surgical field.

TECHNICAL SKILLS

CAD & Design: SOLIDWORKS, Siemens NX, CATIA, Creo, Fusion360, AutoCAD, GD&T, DFM/DFA

Analysis & Simulation: ANSYS, FEA, Vibrational Analysis, MATLAB/Simulink, Python, NIRfast, K-Wave, Systems Engineering (V&V)

Robotics/ML/Controls: ROS 2, Vicon Motion Capture, Autonomous Systems, Linux, MATLAB DL Toolbox, Jupyter Notebook, Git

Manufacturing & Electronics: CNC Lathe/Milling (Sinumerik), Mastercam, Nanofabrication, Arc Welding, FDM (Prusa), MEMS Integration, ESP32, KiCad, Altium Designer, Multisim, C/C++, Embedded Systems (I2C/SPI/UART)